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Attention-guided multimodal skin disease classification using discounted Dempster–Shafer fusion and EfficientNet

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ABSTRACT

Skin diseases represent a significant global health challenge, with skin cancer being the most commonly diagnosed cancer worldwide according to the World Health Organization. Traditional diagnostic methods often rely on visual or biopsy examination, which can be time-consuming and may leave scars on the skin. This approach lacks precision and can delay the initiation of treatment, highlighting the need for alternative diagnostic methods. This study presents a novel multimodal architecture for skin disease classification that combines dermoscopic images and structured clinical metadata. By employing a discounted evidential fusion framework with attention, where each modality independently generates belief mass distributions using an attention-enhanced dense network. These distributions are then adaptively fused using a modified Dempster–Shafer theory, where modality-specific reliability coefficients (α_1, α_2) adjust the weight of each input based on its contextual trustworthiness. The fused beliefs are concatenated with deep image and tabular features for classification, enabling the network to integrate both semantic and probabilistic reasoning. This approach was tested on four benchmark datasets: HAM10000, ISIC 2019, ISIC 2020 and PAD-UFES-20. Results confirm its generalizability and demonstrating the power of combining these two modalities with probabilistic fusion methods to address the challenges inherent in diagnosing skin diseases, paving the way for more effective and efficient healthcare solutions.

1. Introduction

Skin diseases encompass a wide range of disorders from common conditions like acne and inflammation to serious disorders like melanoma [1]. These diseases place a significant burden on global health systems [2], affecting millions of people worldwide and requiring precise, rapid diagnosis for effective treatment [2]. However, classification of skin diseases is inherently difficult due to the great heterogeneity of their presentation: the same disease can take different forms and can have very subtle differences from other conditions [3]. Other factors may influence, including patient demographics and medical history.

Traditional diagnostic methods that dermatologists follow often rely heavily on visual inspection and clinical expertise [4]. Although these methods benefit from the expertise and experience of healthcare professionals, and have long been applied, they are also subject to human error and variability. In addition, the complexity and diversity of skin conditions, as well as the urgency of treating certain lesions, such as cancer, require more sophisticated and faster diagnostic approaches. The increasing prevalence of skin diseases and the limited availability of dermatologists in many regions highlight the need for automated and scalable diagnostic tools [2,5].

In recent years, advances in deep learning and artificial intelligence have shown promise in the field of medical imaging, providing tools capable of surpassing human capabilities in image recognition tasks, and delivering rapid, accurate diagnoses [6]. Convolutional Neural Networks (CNNs), in particular, have demonstrated exceptional performance in image-based classification tasks, including medical image analysis [7]. Among the various CNN architectures, the EfficientNet model has emerged as a state-of-the-art model due to its ability to achieve high accuracy. It is recognized for its robust architecture that

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achieves an optimal balance between accuracy and a relatively low number of parameters, making it both powerful and computationally efficient [8].

However, while deep learning models excel in visual analysis, they often overlook valuable information contained in records and tabular data [9]. Although CNNs have been highly successful in image classification, depending exclusively on visual data may not provide a comprehensive diagnosis. Integrating additional information, such as patient records, medical history, and other clinical parameters, often enhances the classification of skin diseases [10]. Despite this, most research efforts in the domain of skin disease diagnosis have predominantly focused on processing and analyzing images alone. These models aim to leverage the powerful feature extraction capabilities of CNNs to identify and classify dermatological conditions from image data, often overlooking the rich, complementary information available in patient records and other clinical data, which can be crucial for accurate and nuanced diagnoses [11,12].

To address these limitations, a novel multimodal architecture is proposed that fuses EfficientNet-processed images with tabular data using the Dempster–Shafer function. This technique is well suited to managing uncertainty and combining evidence from multiple sources or modalities [13]. In contrast to prior works relying on static or naive fusion, this study introduces an adaptive evidential fusion mechanism where belief mass distributions are computed independently for each modality and then fused based on their reliability levels. The Dempster–Shafer framework is extended with attention-driven belief mass estimation and dynamic trust coefficients α_1 and α_2, allowing the model to contextually calibrate the influence of each data source. The fusion process leverages the strengths of both data modalities, enabling a more comprehensive and robust diagnostic process. The EfficientNet model processes visual information from skin images, while tabular data, which includes patient-specific information and other relevant clinical parameters, is simultaneously leveraged to provide a more holistic view of the patient's condition [8].

The proposed multimodal architecture aims to respond to the different challenges encountered in the classification of skin diseases. It first exploits the powerful feature extraction capabilities of EfficientNet to process complex visual features in skin images, while also incorporating tabular data to enrich the model's understanding and provide context often missed by image-only approaches. Third, the proposed architecture introduces a novel discounted evidential fusion mechanism that not only handles uncertainty but also dynamically adjusts the contribution of each modality according to its contextual reliability, improving robustness in multimodal learning.

Evaluations on four public datasets confirm the superiority of our method over unimodal and early-fusion baselines, demonstrating the effectiveness of our attention-guided belief fusion strategy for interpretable and reliable skin disease classification. The experimental results obtained in the different datasets demonstrate that this new architecture significantly outperforms traditional single-modality approaches, highlighting the potential of multimodal learning in medical diagnosis. Indeed, thanks to the high precision obtained, this approach confirms its robustness and shows promise for developing more effective and accessible diagnostic tools, thus improving outcomes for patients and advancing the field of dermatology.

The main contributions of this work are summarized as follows:

- A novel context-aware multimodal fusion architecture is proposed for skin disease classification, combining dermoscopic images with structured clinical metadata (age, sex, lesion localization) to capture complementary diagnostic information.
- An attention-guided belief mass generation mechanism is implemented for each modality, allowing the model to emphasize salient features and refine class-wise evidential support.
- A discounted evidential fusion strategy is developed, incorporating learned modality-specific trust coefficients (α_1, α_2) to dynamically weight each modality based on reliability. This enhances robustness to uncertain or missing data while enabling explicit representation of epistemic uncertainty.

The sections are organized categorically, presenting the different theoretical and technical aspects of the methodology presented: Section 2 outlines the complexities and challenges associated with skin disease classification, emphasizing the need for accurate and efficient diagnostic tools. Section 3 presents a review of methods and technologies related to the methodology used and addressing the same problem, highlighting their limitations and the need to explore other approaches. The design and implementation of the new multimodal architecture are detailed in Section 4. Section 5 presents and analyzes the experimental results, demonstrating the effectiveness of the proposed approach and comparing it to related works. The limitations of this research as well as the challenges encountered are presented in Section 6, and a summary of the results, as well as a discussion of potential applications and future directions of this approach, is provided in Section 7.

2. Background

This section introduces the key theories and components behind the proposed methodology. A thorough understanding of these elements is essential to grasp the innovative aspects of the proposed multimodal architecture for classifying skin diseases. The process begins with the EfficientNet model, highlighting its architecture and the advantages it offers in terms of accuracy and computational efficiency [8]. Then, we examine the integration of tabular data, which provides additional clinical information thus improving the diagnostic process [9]. Finally, the Dempster–Shafer theory is introduced, emphasizing the merging of the two modalities and the management of uncertainty, along with details of the visual–tabular data fusion process [13]. This theory is extended to introduce a discounted evidential fusion strategy, where each modality generates its own belief mass distribution using attention mechanisms, and their contribution is adaptively calibrated through modality-specific trust factors (α_1, α_2). This approach not only manages uncertainty but also improves robustness by modulating the influence of each modality depending on its reliability, leading to a more context-aware and interpretable decision-making process. The presentation of these theoretical aspects aims to offer a global vision of the approaches used to then detail the implementation and its impact on the diagnosis of skin conditions.

2.1. EfficientNet

EfficientNet is a powerful and widely used convolutional neural network architecture, particularly appreciated for its ability to handle complex image classification tasks with efficiency [8]. Its key innovation lies in its compound scaling strategy: instead of just deepening or widening the network, EfficientNet scales width, depth, and image resolution simultaneously using a unified compound strategy. This thoughtful balance allows the model to reach high accuracy while keeping computational demands low, an essential advantage when dealing with large image datasets. In the medical imaging field, where precision is crucial, EfficientNet has proven to be a reliable choice for extracting meaningful patterns from visual data. Its performance and efficiency make it a strong foundation for integrating image-based information with other clinical inputs in diagnostic systems.

2.2. Dempster–Shafer Theory

The Dempster–Shafer Theory (DST), also known as evidence theory, is a mathematical framework for reasoning under uncertainty, frequently used to represent and combine quantified beliefs [14,15]. It extends Bayesian probability by allowing belief to be assigned to sets or subsets of hypotheses, rather than individual events, thereby capturing both certainty and uncertainty [16].

The core operation in DST is the combination of independent sources of evidence using Dempster's rule of combination. Given two belief functions m_1 and m_2, the combined belief function m is computed as:

$$m(A) = \frac{\sum_{B \cap C = A} m_1(B) \cdot m_2(C)}{1 - \sum_{B \cap C = \emptyset} m_1(B) \cdot m_2(C)} \quad (1)$$

This rule aggregates belief masses where subsets B and C intersect on A, and normalizes by the degree of conflict between the evidence sources.

2.3. Discounted Dempster–Shafer Fusion with adaptive trust modeling

In the proposed architecture, traditional DST is extended through the introduction of a discounted evidential fusion mechanism specifically tailored for multimodal data involving image and tabular clinical inputs. Rather than applying Dempster's rule directly on static belief masses, belief distributions are generated independently for each modality by means of an attention-based network trained to highlight salient features.

A custom Lambda layer in TensorFlow/Keras implements the belief fusion. The key innovation lies in the introduction of learned reliability coefficients, denoted as α_1 and α_2, which are adaptively updated during training rather than fixed.

In Fig. 1, 'belief1' and 'belief2' denote the attention-refined belief mass vectors from the image and tabular branches, respectively. Each belief is dynamically adjusted using a learned trust coefficient α, predicted through a dedicated dense layer, which captures the reliability of the corresponding modality for a given input. To reflect uncertainty, the discounted portion (1 - α) of each belief is replaced with a uniform distribution over all classes, representing epistemic ignorance.

This adjustment ensures that the system remains confident only when warranted by the data and avoids overconfident predictions when input signals are weak or ambiguous. The refined belief vectors are then fused using a normalized Dempster–Shafer combination rule that accounts for both agreement and conflict between modalities. This fusion mechanism enhances the model's robustness to degraded or missing data and improves interpretability by explicitly representing uncertainty and confidence, which is especially valuable in clinical contexts.

3. Related work

Due to the potential to improve diagnostic accuracy and performance, machine learning and artificial intelligence methods for skin disease categorization have attracted much attention in the research literature, according to the survey study by Litjens et al. [7]. Researchers have proposed several approaches using both image and tabular data via multimodal models. This section reviews the main studies related to this research topic and compares them with the proposed approach.

In 2017, Esteva et al. [11] presented a deep learning system for skin cancer image classification that used both medical image data and additional patient information, such as age data as well as biological sex and lesion site details. The system applied convolutional neural network technology to handle image processing while a metadata integration system improved the prediction accuracy. The research results revealed that the model performance improved with metadata integration highlighting the importance of using multiple data sources. The system achieved performance comparable to that of medical experts across two different critical tasks: keratinocyte carcinomas versus benign seborrheic keratoses; and malignant melanomas versus benign nevi. The fusion approach used showed a simple operation mechanism but displayed poor ability to handle uncertain conditions produced by different data sources.

Codella et al. [17] proposed in 2019 a hybrid model that combines dermoscopic images with patient data and genetic information. Their approach was based on two separate neural networks employed for each data type and then their outputs were fused at the feature level. When applied to the ISBI2016 Dataset, it achieves a sensitivity rate of 95%. The research demonstrated how using multiple data sources worked yet noted difficulties in controlling the contribution of each modality especially when the data was noisy or incomplete.

In contrast to these simpler approaches, a more sophisticated strategy was described by Gessert et al. [18] who used a late fusion multi-modal approach to combine CNNs together with patient metadata. In this method, image and tabular data streams were processed separately before being combined in the final classification layer using a late fusion technique. Applied on ISIC 2019, it achieves a balanced accuracy rate of 74.2%. Although this approach leveraged both data types effectively, the late fusion technique failed to maximize the potential interdependencies of the modalities.

In another research, Tajjour et al. [19] presented a novel combination of a Convolutional Neural Network (CNN) and a Multilayer Perceptron (MLP) for the early detection of skin cancer. This approach also included both image data and manually extracted structured data (patient data) which was fed into the model after applying several color space conversions for lesion detection. Applied to the HAM10000 dataset, the hybrid model achieved 86.0% top-1 accuracy, 95.0% top-2 accuracy, and an AUC of 0.96. The hybrid model showed a 2% improvement in accuracy as compared to the standalone model, making it suitable for handling both the structured and unstructured data. A major limitation of this approach is that the method relies on manual feature extraction which can lead to bias and may decrease the level of automation as compared to fully end-to-end deep learning methods.

A further improvement in multi-modal ensemble strategies was presented by Sharafudeen et al. in [20], where the authors proposed an integrated multi-modal ensemble framework that combines deep convolutional neural representations with extracted lesion characteristics and patient metadata. The approach combines transfer learned image features with global and local textural information and patient data using a custom generator for skin cancer diagnosis. The model is a weighted ensemble strategy which is trained and validated on HAM10000, BCN20000 + MSK, and ISIC2020 challenge datasets, which gives sensitivity scores of 94.15%, 86.69%, and 86.48%, and specificity scores of 99.24%, 97.73%, and 98.51, respectively. Furthermore, accuracy on the malignant classes was 94%, 87.33%, and 89%, which is greater than that of physician recognition rates. However, this ensemble strategy, as promising as it is, may increase the computational complexity and need for data preprocessing, which makes real-time implementation more difficult.

Zhou and Luo presented significant contribution in their paper [21] which implements a deep learning framework that uses multimodal data for skin lesion diagnosis. In their end-to-end neural network model, they combined dermoscopic lesion images with textual information inspired by vision-language models in Visual Question Answering (VQA). The Mutual Attention Transformer (MAT) which they developed consists of self-attention and guided-attention blocks to enable interaction between the two modalities. The fusion mechanism added the features extracted before classification which resulted in a performance of 92.55% accuracy and 98.28% AUC on the HAM10000 dataset. The limitation of this method is that it is computationally expensive due to the complexity of transformer-based architectures and requires extensive fine-tuning to effectively balance multimodal features.

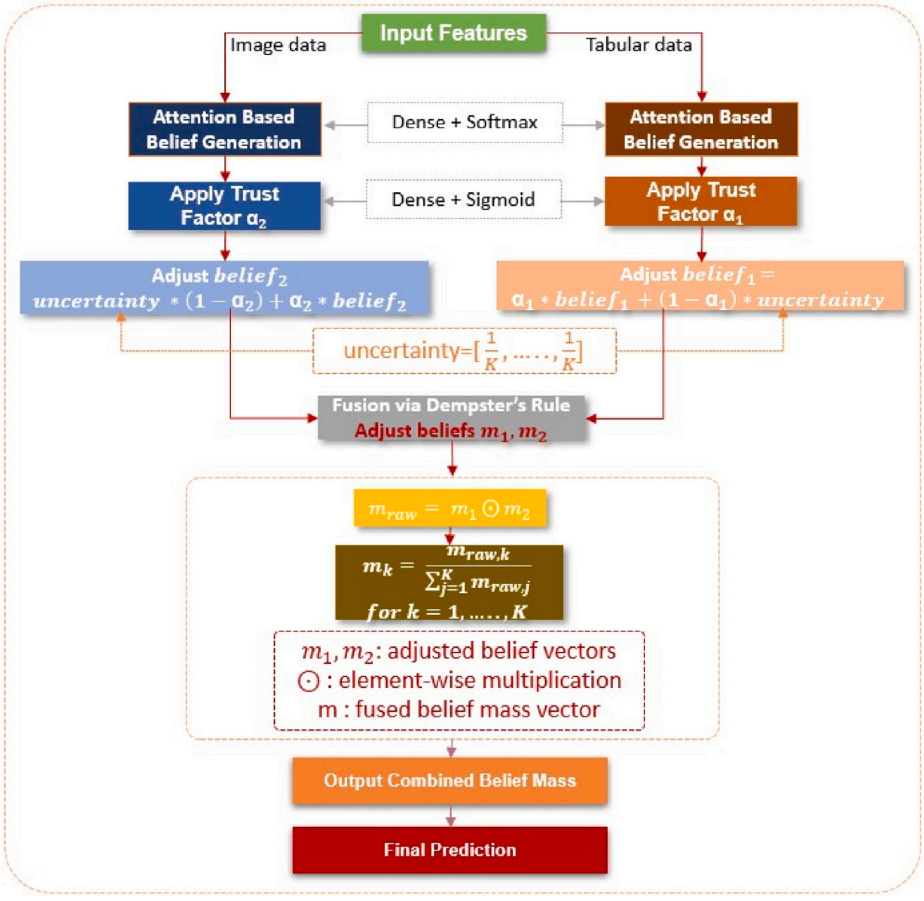


Fig. 1. Flowchart of the discounted evidential fusion process.

In their paper, Khan et al. [22] proposed an automated skin cancer diagnosis and triaging model that can improve diagnostic accuracy through clinical features. An ensemble learning framework was used by the study to combine EfficientNetB3 for analysis of skin lesions with Extreme Gradient Boosting (XGB) for classification of clinical data. To improve data class imbalance, PAD-UFES-20 dataset with six unbalanced skin cancer categories was subjected to data augmentation. The model showed better performance compared to previous studies with an accuracy of 0.78, precision of 0.89, recall of 0.86 and F1 score of 0.88. The use of an ensemble approach increases the computational cost and needs a lot of hyperparameter tuning.

Similarly with our research, Sarkar et al. [23] proposed a skin cancer classification model that uses Dempster–Shafer theory to combine the outputs of three attention based CNNs that work separately. Their model uses ResNet50, InceptionResNetV2, and DenseNet201, all with a soft attention module, which achieves 93.2% accuracy on HAM10000 dataset. However, their approach is to fuse multiple CNN architectures into one model while our research is to apply Dempster–Shafer theory to integrate multimodal data, including structured metadata along with unstructured image data for improved diagnostic accuracy. The main disadvantage is that the use of multiple deep learning models is a computationally expensive and time–consuming process that makes it difficult to implement the model in real time.

The proposed study develops a novel approach by pairing EfficientNet with tabular data within a multi-modal architecture while addressing some limitations which previous methods faced. The EfficientNet model balances accuracy and computational cost more effectively than traditional CNNs. Also, the Dempster–Shafer theory brings a significant enhancement to previous data fusion approaches. This probabilistic method to combine evidence helps to handle uncertainty and, therefore, improves the use of multi-modal data.

As described in Table 1, this paper combines the EfficientNet model with a probabilistic fusion technique based on Dempster–Shafer theory which distinguishes it from previous works and provides a better and more efficient solution for the task of skin disease classification. Gessert et al. [18] used the late fusion approach which did not properly utilize the relations between image and metadata modalities. The proposed model applies the Dempster–Shafer theory to perform probabilistic fusion, thus providing a better way of fusing complementary data sources. Furthermore, Codella et al. [17] showed that the combination of multiple data types was effective, but they encountered difficulties in how much each modality should contribute. This problem is addressed by using a structured fusion framework that adaptively sets weights according to the uncertainty of evidence.

Tajjour et al. [19] added structured features that were beneficial but also brought bias and decreased automation. Our model avoids manual feature engineering by using deep learning to derive important representations from tabular metadata and images, thus building a more solid and extensible solution. Nevertheless, Khan et al. [22] achieved better diagnostic accuracy through an ensemble method at the expense of higher computational cost and many hyperparameters to tune. Our method does this by choosing EfficientNet which offers high accuracy with low computational requirements. Zhou and Luo [21] employed a Mutual Attention Transformer (MAT) for multimodal fusion but needed much fine-tuning and computational resources. Instead, our approach uses a simpler yet equally effective fusion method which produces similar results to transformer-based models without their excessive requirements. Finally, Sarkar et al. [23] applied Dempster–Shafer theory for fusion of CNN models, however, our study builds

Table 1
Comparative study of multi-modal approaches in skin disease classification.

Study	Data types used	Methodology	Key findings
Esteva et al. [11]	Image and metadata	Applied CNNs with metadata integration.	Improved prediction accuracy but struggled with uncertainty in heterogeneous data sources.
Codella et al. [17]	Image, patient data, and genetic information	Hybrid model using separate networks for each modality, fused at the feature level.	Achieved high sensitivity but faced challenges in balancing contributions from different modalities.
Gessert et al. [18]	Image and patient metadata	Used a late fusion strategy combining CNN outputs and metadata.	Showed effectiveness but lacked optimal exploitation of interdependencies between data types.
Tajjour et al. [19]	Image and structured data	Combined CNN with MLP and manually extracted features from color space conversions.	Improved accuracy but relied on manual feature extraction, reducing automation.
Sharafudeen et al. [20]	Image and lesion characteristics	Integrated deep convolutional features with extracted lesion metadata using a weighted ensemble.	Achieved high sensitivity and specificity but increased computational complexity.
Zhou and Luo [21]	Image and textual data	Developed Mutual Attention Transformer (MAT) for vision-language fusion.	Achieved high accuracy but required extensive fine-tuning and high computational resources.
Khan et al. [22]	Image and clinical data	Combined EfficientNetB3 for image analysis with XGB for metadata classification.	Enhanced diagnostic accuracy but increased computational cost and hyperparameter tuning complexity.
Sarkar et al. [23]	Image	Combined multiple attention-based CNN models using Dempster–Shafer theory.	Improved classification accuracy but had high computational cost.
Current Study	Image and tabular data	Integrates EfficientNet for image analysis with tabular data using the Dempster–Shafer fusion technique.	Addresses limitations of prior methods, providing a more effective and efficient solution for skin disease classification by considering interdependencies and managing uncertainty.

upon this idea to perform multimodal fusion of structured patient metadata with imaging data which leads to better diagnostic accuracy. As for these limitations, the proposed approach gives a more efficient, more scalable, and more clinically relevant solution for skin disease classification.

In fact, this study differs from prior works by introducing a novel adaptive evidential fusion mechanism. Unlike existing approaches that rely on early or basic fusion strategies, the proposed method generates attention-guided belief mass vectors for each modality and dynamically adjusts their contribution using modality-specific reliability factors (α_1, α_2). The combination is performed using an extended Dempster–Shafer rule, enabling uncertainty-aware integration of visual and tabular information. This formulation results in notable improvements in both accuracy and robustness, and demonstrates the potential of trust-weighted multi-modal architectures in clinical diagnostic tasks.

4. Methodology

This section outlines the methodology used in this study, including the workflow, data sources, preprocessing, and training strategies. The proposed framework combines visual and tabular modalities to enable robust skin disease classification. A brief overview of the model pipeline is presented, followed by the integration approach, dataset details, and implementation settings.

4.1. Workflow

The proposed workflow follows a hybrid multimodal architecture that processes dermoscopic images and structured clinical metadata in parallel. Image features are extracted using the EfficientNet backbone, while clinical attributes such as age, sex, and lesion localization are encoded through a dedicated dense network.

Belief mass vectors are generated for each modality and combined using a discounted Dempster–Shafer rule, integrating information under uncertainty. The resulting fused belief mass is concatenated with high-level features and passed through dense layers for final classification, as illustrated in Fig. 2.

To facilitate convergence, the training process employs a warm-up phase followed by a cosine learning rate scheduler. Learning rate

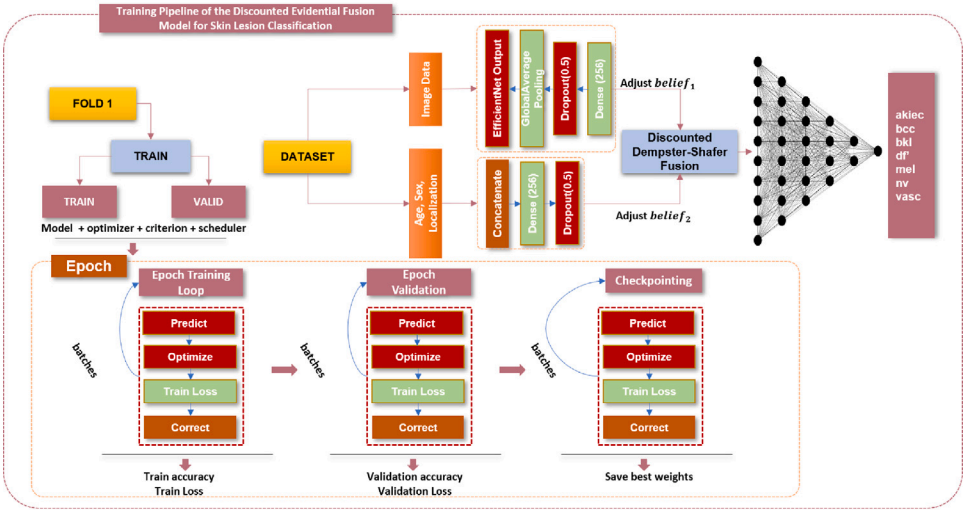


Fig. 2. General workflow of the proposed fusion-based architecture.

callbacks further support dynamic adaptation throughout training, contributing to stable optimization and improved generalization.

4.2. Dempster–Shafer and multi-modal skin disease classification

The proposed approach brings together dermoscopic images and clinical metadata by encoding each modality into belief mass distributions. Their combination is achieved through a discounted evidential fusion mechanism grounded in Dempster–Shafer Theory, allowing uncertainty to be effectively handled. Trust factors are learned to dynamically adjust the reliability assigned to each data source, ensuring that the model emphasizes the most informative modality in each case. These trust weights are not fixed; they are continuously updated during training as the model observes more examples and learns to assess the quality of each modality in varying clinical contexts. For instance, when metadata is missing or noisy, the model learns to reduce reliance on the tabular stream and prioritize image features; when image quality is poor, it relies more on metadata. As a result, predictions are made with improved confidence and contextual awareness.

Let I denote the image input and T the tabular clinical input. Feature vectors are first extracted using a convolutional neural network (CNN) and a multi-layer perceptron (MLP):

$$\mathbf{f}_I = f_{\text{CNN}}(I;\theta_I), \quad \mathbf{f}_T = f_{\text{MLP}}(T;\theta_T) \tag{2}$$

These feature vectors are then mapped to belief mass vectors m_1 and m_2, respectively:

$$m_1 = \phi_I(\mathbf{f}_I), \quad m_2 = \phi_T(\mathbf{f}_T) \tag{3}$$

To model modality reliability, each belief is adjusted using a trust coefficient $\alpha \in [0,1]$, and a uniform distribution $u = \left[\frac{1}{K},\ldots,\frac{1}{K}\right]$ over K classes, representing complete uncertainty. The trust coefficients α_1 and α_2 are learned through dedicated dense layers with sigmoid activation, applied to the image and tabular feature vectors respectively. This allows the network to dynamically assess the reliability of each modality in a data-dependent manner:

$$\tilde{m}_1 = \alpha_1 \cdot m_1 + (1-\alpha_1)\cdot u, \quad \tilde{m}_2 = \alpha_2 \cdot m_2 + (1-\alpha_2)\cdot u \tag{4}$$

These adjusted beliefs are then combined using Dempster's rule of combination:

$$m(A) = \frac{\sum_{B\cap C=A} \tilde{m}_1(B)\cdot\tilde{m}_2(C)}{1-\sum_{B\cap C=\emptyset}\tilde{m}_1(B)\cdot\tilde{m}_2(C)} \tag{5}$$

The resulting belief mass m represents the final evidence distribution across diagnostic classes. The model is trained to maximize the belief assigned to the true class A_{true} using the following objective:

$$\mathcal{L} = -\log m(A_{\text{true}}) \tag{6}$$

This architecture enables adaptive fusion of modalities by adjusting belief reliability dynamically during training. It is particularly effective for integrating uncertain or heterogeneous medical data, as it provides a principled way to combine probabilistic evidence while accounting for partial knowledge or modality conflict.

4.3. Dataset

- HAM10000 (Human Against Machine): This diverse dataset contains 10,015 dermoscopic images representing skin lesions across seven diagnostic categories: actinic keratosis (AKIEC), basal cell carcinoma (BCC), benign keratosis (BKL), dermatofibroma (DF), melanoma (MEL), melanocytic nevus (NV), and vascular lesions (VASC) [24].
- ISIC 2019: A large-scale dataset with over 25,000 images spanning eight disease classes (MEL (Melanoma), NV (Melanocytic Nevus), BCC (Basal Cell Carcinoma), AK (Actinic Keratosis), BKL (Benign Keratosis-like Lesions), DF (Dermatofibroma), VASC (Vascular Lesions), and SCC (Squamous Cell Carcinoma)). This dataset has a more imbalanced class distribution, making it challenging for classifiers.
- ISIC 2020: Part of the International Skin Imaging Collaboration (ISIC) challenge, consisting of 2,000 images with annotations to distinguish between benign and malignant skin lesions [25].
- PAD-UFES-20: This dataset comprises 2,298 clinical images of six types of skin lesions (NEV (Nevus), SCC (Squamous Cell Carcinoma), BCC (Basal Cell Carcinoma), ACK (Actinic Keratosis), SEK (Seborrheic Keratosis), and MEL (Melanoma)), including metadata such as patient age, lesion location, and skin type. Collected by the Dermatological and Surgical Assistance Program at the Federal University of Espírito Santo, it aims to support research in skin lesion analysis [26]. For analysis, non-melanocytic skin cancers (SCC, BCC, ACK) were grouped under 'Non-Melanocytic Lesions', while melanocytic lesions (SEK, MEL) were grouped under 'Melanocytic Lesions.' NEV was treated separately as benign nevi.

For the clinical metadata, the tabular data (age, sex, and lesion location) was preprocessed to ensure compatibility with the visual

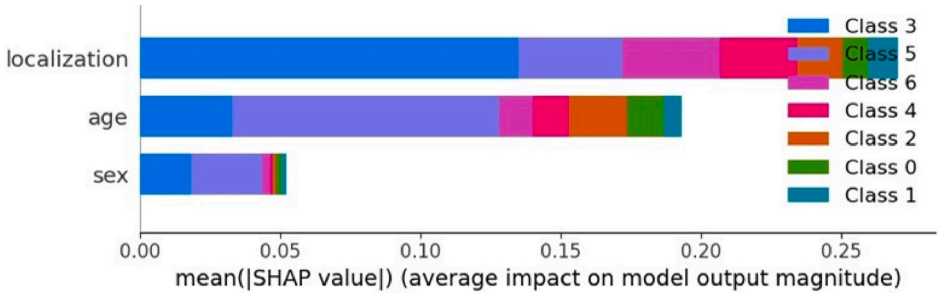


Fig. 3. SHAP feature importance analysis for tabular data, showing the average impact of each feature on model predictions across different classes.

data. Missing values in the `age_approx` column were imputed with the mean age, while `anatom_site_general_challenge` entries were set to 'Unknown'. The `sex` column was completed by replacing missing entries with the most frequent value (mode). Continuous features, such as age, were scaled to the [0, 1] range using MinMaxScaler to ensure uniformity, and categorical features, such as `sex` and `anatom_site_general_challenge`, were label-encoded into numerical values. This preprocessing step facilitated the seamless integration of clinical data with features extracted from the dermoscopy images.

In this experiment, the datasets were randomly divided into a training, validation, and test set in a 80:10:10.

4.4. Clinical data preprocessing

To ensure the clinical data was appropriately prepared and integrated with the images, a preprocessing pipeline was applied as described in the Dataset section. The images underwent augmentation using ImageDataGenerator, applying techniques such as rescaling pixel values to [0, 1], rotation, translation, zoom, shearing, horizontal flipping, and brightness variation. These augmentations were applied only to the training set to improve generalization.

To integrate the clinical data with visual features extracted by EfficientNet, the tabular data was indexed and aligned with image filenames. A custom generator was created to load and preprocess both data types simultaneously. The metadata entries were ordered based on dataset splits (train, test) and image identifiers to ensure proper alignment. For each sample, clinical features (age, sex, and lesion location) were concatenated with image feature embeddings from EfficientNet during training. This multimodal approach enabled the model to leverage complementary information from clinical and visual domains.

Fig. 3 illustrates the SHAP (SHapley Additive exPlanations) analysis of the tabular data features (localization, age, and sex) across different classes. The feature "localization" exhibits the highest average SHAP value, indicating its significant contribution to the model's predictions for certain classes, especially Class 3 and Class 5. Age also shows a substantial impact, while sex has the least influence on the model's output. This analysis highlights the importance of integrating clinical metadata to enhance the model's interpretability and diagnostic performance.

4.5. Data augmentation

The purpose of data augmentation is to expand the dataset size by transforming the existing data [27]. The following transformations were applied: horizontal and vertical flipping, rotation, and color normalization, among others.

As noted by Naik et al. [28], the color distribution differed based on the lighting conditions, dermatoscope types and other factors. These differences may negatively affect the results and is therefore regarded as noise. Therefore, a color normalization phase was applied where all three channels were averaged and then normalized across the entire dataset. The "Shades-of-Grey" algorithm [29] was also applied which normalizes the color in images to reduce the variance that comes from different imaging conditions and devices [30].

5. Experimental results and discussion

To evaluate the performance of the proposed methodology, the proposed architecture was tested on four common dermatological datasets: HAM10000, ISIC2019, ISIC2020 and PAD-UFES-20. The method was implemented using the Tensorflow and Keras frameworks and the experiments were conducted with a GeForce RTX 2060 GPU that has 6 GB GDDR5 memory and the system has 16.0 GB of RAM running Windows10. The training of the model is designed to optimize performance. Categorical cross-entropy was used as the loss function, and the Adam optimizer was selected for its adaptive learning rate, which helps in converging the network faster. A dynamic learning rate scheduler is used to change the learning rate of an optimizer during training as a function of the current training progress, decreasing the learning rate when the validation loss is plateauing to improve the model's accuracy without overfitting.

Each dataset presents unique characteristics and challenges, such as varying class distributions and image qualities, providing a comprehensive evaluation of the model's ability to handle diverse real-world scenarios.

The evaluation metrics include accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC-ROC). Additionally, the training curves, confusion matrices, and ROC curves for each dataset are provided to analyze the model's learning behavior, classification performance, and decision-making reliability.

5.1. Ablation study

The ablation study aims to evaluate the contribution of each key component of the proposed model to its overall performance. The model was trained with a batch size carefully chosen to balance computational efficiency and performance. Training was conducted until convergence, monitored by early stopping criteria based on validation loss to prevent overtraining. The evaluation metrics used were accuracy, precision, recall, and F1-score, providing a comprehensive assessment of the model's performance under different configurations. To highlight the impact of each component, we tested the model in four scenarios: (1) using the EfficientNet backbone alone (Baseline: images only), (2) using only the tabular clinical data, (3) combining visual and tabular modalities with a simple concatenation-based fusion, and (4) the full proposed model with multi-modal fusion via the Dempster–Shafer function. The ablation study presented in Table 2 evaluates the contributions of each component in the proposed model on the HAM10000 dataset.

The results demonstrate that EfficientNet, using image data alone, achieves high accuracy (92.5%) with strong precision (93%) and recall (92%). However, XGBoost, leveraging only tabular data, underperforms with an accuracy of 81.3%, highlighting the limited discriminative power of metadata in isolation. Simple fusion through concatenation improves accuracy to 93.5%, indicating that metadata provides complementary insights when integrated with image features. The proposed model, utilizing advanced fusion mechanisms, achieves

Table 2
Ablation study results on the HAM10000 dataset.

Configuration	Accuracy	Precision	Recall	F1-score
EfficientNet (Images Only)	92.5%	93%	92%	91%
XGBoost (Tabular Data)	81.3%	–	–	–
Fusion With Concatenation	93.5%	93.5%	93.3%	93%
Proposed Model (Full Fusion)	**94.8%**	**94.78%**	**94.82%**	**95%**

Table 3
Performance metrics of the proposed model on HAM10000, ISIC 2019, ISIC 2020, and PAD-UFES-20 datasets.

Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
HAM10000	94.8	94.78	94.82	95
ISIC 2019	84.67	84.3	84.7	85
ISIC 2020	98.57	98.78	98.18	99
PAD-UFES-20	90.87	90.52	90.87	90

the best performance with an accuracy of 94.8%, precision of 94.78%, recall of 94.82%, and an F1-score of 95%. This highlights the efficacy of sophisticated fusion techniques in leveraging multi-modal data for improved classification, demonstrating a balanced and robust performance suitable for clinical applications.

5.2. Evaluation on multiple datasets

To evaluate the generalization and robustness of the proposed model, we tested its performance across three well-established datasets: HAM10000, ISIC 2019, ISIC 2020, and PAD-UFES-20.

5.2.1. Performance metrics

Table 3 summarizes the classification performance of the proposed model across the three datasets.

The proposed model achieves outstanding performance on the HAM10000 dataset, with an accuracy of 94.8%, precision of 94.78%, recall of 94.82%, and an F1-score of 95%. The results show that the model can achieve a good trade-off between sensitivity and specificity and thus make correct predictions across different skin lesion classes.

On the ISIC 2019 dataset the model has an accuracy of 84.67%, precision of 84.3%, recall of 84.7% and F1 score of 85%. Although the performance is slightly lower than that obtained with HAM10000, it can be attributed to the higher complexity and class imbalance of the ISIC 2019 dataset, which is more difficult to learn for classification models.

On the ISIC 2020 dataset the model has an accuracy of 98.57%, precision of 98.78%, recall of 98.18% and F1 score of 99%. These results show that the model generalizes well to high quality datasets and thus it is very relevant for application in real clinical settings.

On the PAD-UFES-20 dataset the model has an accuracy of 90.87%, precision of 90.52%, recall of 90.87% and F1 score of 90%. These results show that the model is strong in handling a challenging dataset with grouped dermatological categories and can distinguish between benign and malignant skin lesions in different clinical settings.

Training and validation curves

Fig. 4, 5, 6 and 7 show the training and validation accuracy and loss curves for the HAM10000, ISIC 2019, ISIC 2020, and PAD-UFES-20 datasets. The model achieves smooth convergence across all datasets, with minimal overfitting, as indicated by the alignment of training and validation curves.

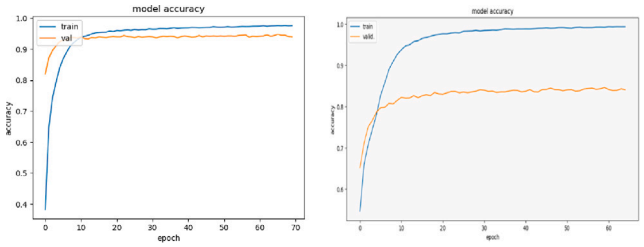


Fig. 4. Training and validation accuracy curves for HAM10000 (left) and ISIC 2019 (right).

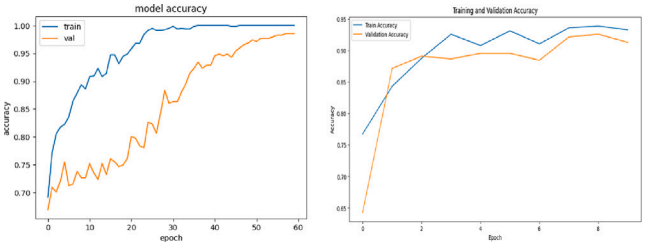


Fig. 5. Training and validation accuracy curves for ISIC 2020 (left) and PAD-UFES-20 (right).

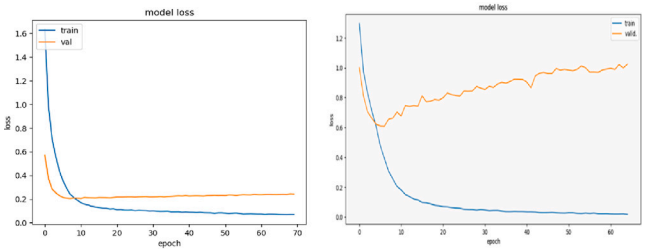


Fig. 6. Training and validation loss curves for HAM10000 (left) and ISIC 2019 (right).

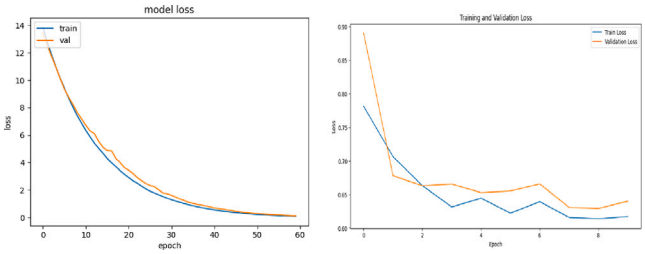


Fig. 7. Training and validation loss curves for ISIC 2020 (left) and PAD-UFES-20 (right).

5.2.2. Confusion matrices

Figs. 8 and 9 display the confusion matrices for the four datasets. The confusion matrices highlight the balance in classification across classes, with fewer misclassifications on the ISIC 2020 dataset, which achieves the highest performance.

The model demonstrates strong classification performance across the HAM10000 dataset, with the highest accuracy observed for the "nv" and "vasc" classes. However, there are some notable misclassifications. The "df" class is often mistaken for "akiec" or "mel" attributes which share similarities with this class. The "mel" class is also mistaken for "nv" which is understandable because both classes are similar in appearance. Though the overall accuracy is very high, these findings suggest that there is scope for further improvement to enhance the model's specificity to distinguish between visually similar classes.

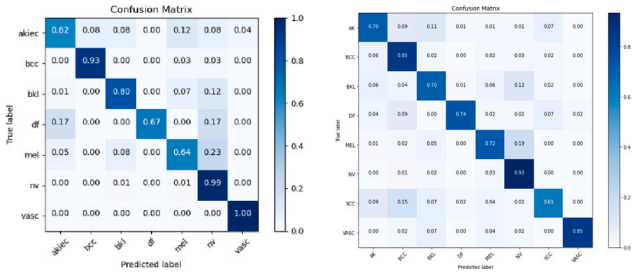


Fig. 8. Confusion matrices for HAM10000 (left) and ISIC 2019 (right).

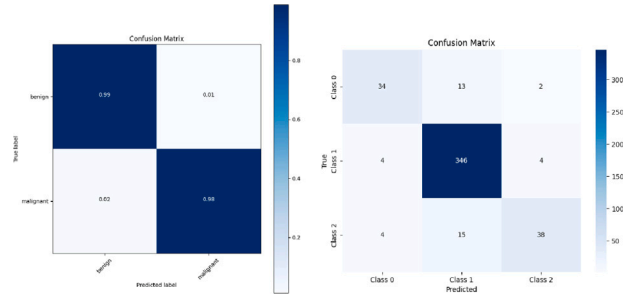


Fig. 9. Confusion matrices for ISIC 2020 (left) and PAD-UFES-20 (right).

The model shows a slight deterioration in performance when used with the ISIC 2019 dataset as compared to HAM10000. This is especially so for the "scc" and "df" classes which have a relatively high rate of misclassification. The "bkl" class is also difficult to predict as many of the samples are misclassified as either "nv" or "mel". These results indicate that the complexity and the imbalanced nature of the ISIC 2019 dataset affect the model's effectiveness in identifying minority classes accurately.

On the ISIC 2020 dataset the model has a nearly perfect classification of the data with 99% accuracy for the benign class and 98% for the malignant class. These two classes have very few misclassifications, which shows that the model has a good generalization ability. The high quality and balanced nature of the ISIC 2020 dataset may be attributed to this superior performance as the model is very robust in well-curated datasets.

From the confusion matrix, it is evident that the model has excellent classification accuracy across the three groups of classes: Benign Nevi (Class 0), Non-Melanocytic Lesions (Class 1), and Melanocytic Lesions (Class 2) in the PAD-UFES-20 dataset. The model makes a very large number of correct predictions in Class 1 with only 346 incorrect predictions, indicating its high accuracy in detecting non-melanocytic lesions. The confusion matrices show that the model performs well across all the four datasets, with the best performance seen in the ISIC 2020 dataset. The performance drop on ISIC 2019 is due to issues related to dataset imbalance and complexity that lead to increased misclassification rates in certain minority classes.

All four datasets show the model's robustness through confusion matrices with the strongest results seen at ISIC 2020. The performance drop on ISIC 2019 reveals the challenges of dataset imbalance and complexity resulting in higher misclassification rates among minority classes.

5.2.3. ROC curves

Figs. 10, 11, 12 and 13 illustrate the ROC curves for the HAM10000, ISIC 2019, ISIC 2020, and PAD-UFES-20 datasets. The model achieves the highest AUC on the ISIC 2020 dataset, indicating excellent separability between classes. The ROC curve for ISIC 2019 shows slightly reduced performance, consistent with its lower accuracy and F1-score.

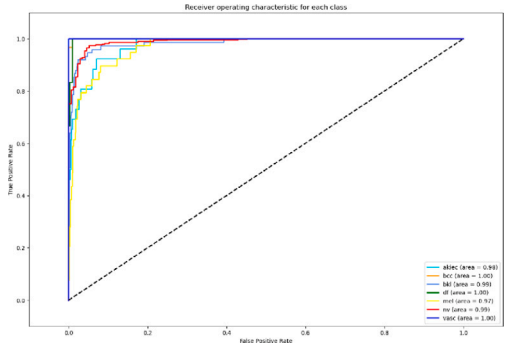


Fig. 10. ROC curve for the HAM10000 dataset.

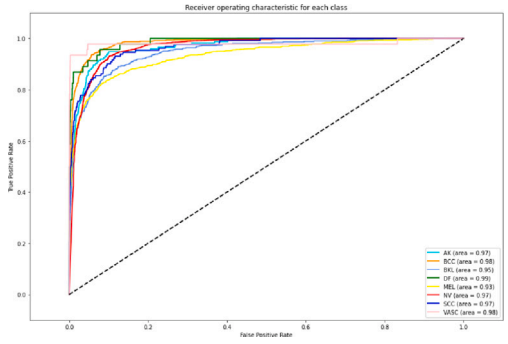


Fig. 11. ROC curve for the ISIC 2019 dataset.

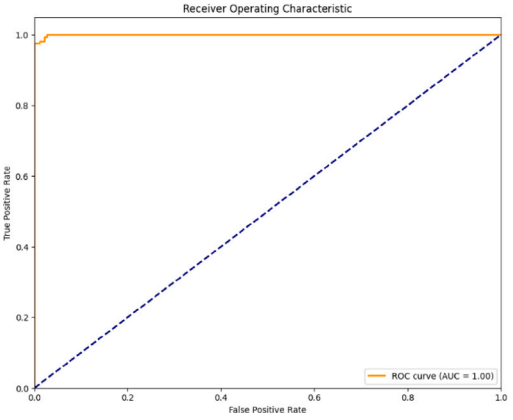


Fig. 12. ROC curve for the ISIC 2020 dataset.

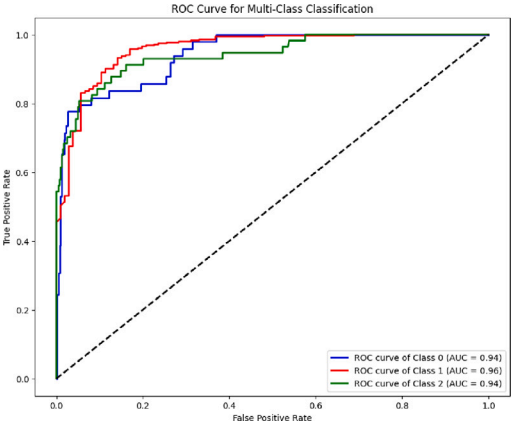


Fig. 13. ROC curve for the PAD-UFES-20 dataset.

The model performs excellently on HAM10000 dataset with AUC values of 0.98 or higher for most classes and perfect classification for classes such as “bcc” and “df”. The performance for ISIC 2019 is strong with AUC values above 0.93 across all classes but lower for “mel” and “bkl” classes because of the higher complexity of the dataset. The ISIC 2020 dataset shows near perfect classification with an AUC of 1.0, which suggests that the model is very effective in distinguishing between “benign” and “malignant” cases in a balanced dataset. These results show that the model has strong generalization performance across different datasets, with its strongest performance observed on ISIC 2020. For the PAD-UFES-20 dataset, the ROC curves demonstrate strong classification performance across the three classes: Benign Nevi (Class 0), Non-Melanocytic Lesions (Class 1), and Melanocytic Lesions (Class 2), with AUC values of 0.94, 0.96, and 0.94 respectively. These high AUC values indicate excellent discriminative ability of the model in distinguishing between the classes, reflecting robust predictive accuracy.

5.3. Interpretable fusion using Dempster–Shafer

The Dempster–Shafer theory provides a robust framework for combining evidence from multiple modalities while accounting for uncertainty and conflict in the data. To ensure clinical applicability, it is crucial to provide transparency in the fusion process, enabling clinicians to interpret and trust the model’s decisions. Interpretability can be enhanced by leveraging SHAP to analyze feature contributions from both tabular and image data [31].

- **Feature Importance for Tabular Data:** Using SHAP, the contributions of individual clinical features, such as age, sex, and lesion location, can be quantified [32]. For instance, SHAP values could reveal that age significantly influences a melanoma diagnosis, while sex has a lower impact. These insights allow clinicians to understand the role of each feature in the model’s decisions.
- **Image Feature Interpretability:** LIME (Local Interpretable Model-Agnostic Explanations) [33] and Grad-Cam (Visual Explanations from Deep Networks via Gradient-based Localization) [34] can be applied to analyze the feature vectors extracted from the CNN branch. By mapping LIME explanations to specific image features, clinicians can understand which aspects of the image (e.g., texture or color patterns) contribute most to the model’s belief mass for a specific diagnosis.

Fig. 14 presents LIME visual explanations for randomly selected dermoscopic images, showcasing the regions most influential in the model’s predictions. The highlighted regions (in yellow and red) indicate the areas that contributed most to the classification decision. For instance, in the examples, the model correctly identified lesions by focusing on distinct regions associated with relevant features. These visualizations enhance the interpretability of the model, providing insights into the reasoning behind its predictions, which is critical for clinical decision-making. In addition to LIME, Grad-CAM is also used to assess the model’s focus and interpretability in automated skin disease classification. Fig. 15 presents the original dermoscopic images (left), attention heatmaps highlighting the most influential regions (middle), and the overlay of heatmaps on the original images (right).

5.4. Comparison with state-of-the-art

To further validate the effectiveness of the proposed model, we compare its performance with state-of-the-art methods reported in the literature on the same datasets. These methods represent established benchmarks for skin disease classification tasks and include both single-modal and multi-modal approaches.

Table 4 shows the performance of the proposed model compared to several state-of-the-art methods on different skin diseases datasets.

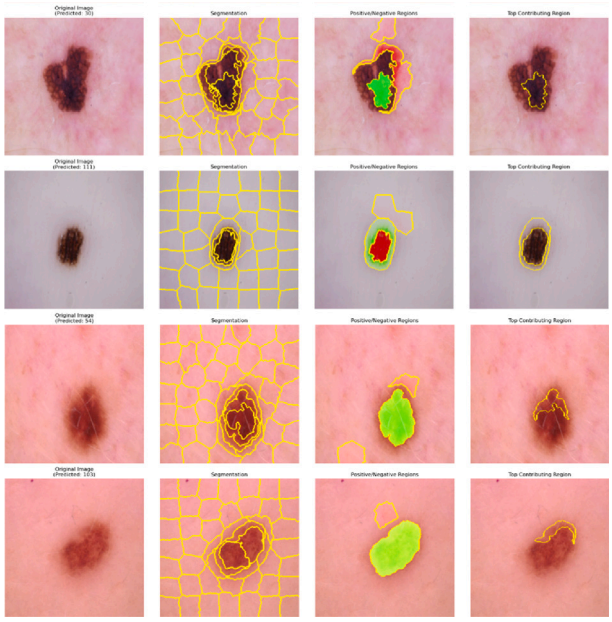


Fig. 14. LIME explanations for dermoscopic images, highlighting the most influential regions contributing to the model’s predictions for different lesion classes.

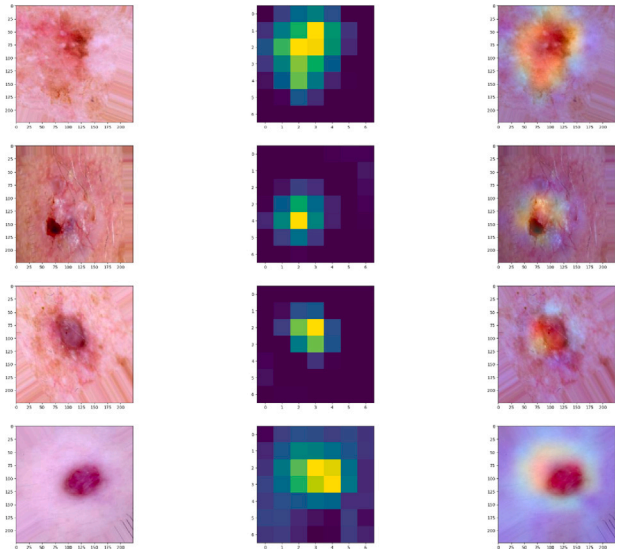


Fig. 15. Visualization of skin lesion classification. The left column shows the original images, the middle column represents attention heatmaps, and the right column overlays the heatmaps on the original images.

The results show that the proposed model has the best performance on the ISIC 2020 dataset with an accuracy of 98.57% and an AUC of 1.00, which is greater than other approaches like Misbah et al. [36] who obtained an accuracy rate of 0.945. On the PAD-UFES-20 dataset, our model also outperforms Ou et al. [35] achieving both higher accuracy (90.87% vs. 76.8%) and AUC (0.95 vs. 0.947). This result shows the effectiveness of the proposed multimodal fusion strategy in dealing with datasets with a limited number of classes.

Several models were evaluated on the HAM10000 dataset, providing a strong benchmark for performance comparison. The proposed model achieves the highest accuracy at 94.8%, along with a balance between precision (94.78%) and recall (94.82%), confirming its strong and consistent performance across classes. Compared to Sharafudeen

Table 4
Comparison of model performance on skin lesion datasets.

Model	Dataset	Accuracy	Precision	Recall	AUC
Esteva et al. [11]	Clinical Dataset	72.1%	–	–	0.96
Codella et al. [17]	ISBI 2016	85%	50.7%	94.1%	0.804
Gessert et al. [18]	ISIC 2019	74.2%	–	74.2%	–
Tajjour et al. [19]	HAM10000	86.0%	–	–	0.96
Sharafudeen et al. [20]	HAM10000	94%	99.24%	94.14%	0.982
Zhou and Luo [21]	HAM10000	92.55%	–	–	0.982
Khan et al. [22]	PAD-UFES-20	78.0%	89.0%	86.0%	–
Sarkar et al. [23]	HAM10000	93.2%	93.15%	93.2%	–
Ou et al. [35]	PAD-UFES-20	76.8%	78.11%	77.51%	0.947
Misbah et al. [36]	ISIC 2020	94.5%	85.4%	84.3%	–
Vivek and Ramesh [37]	HAM10000	87.5%	87.5%	87.5%	0.83
Proposed Model	HAM10000	94.8%	94.78%	94.82%	0.98
Proposed Model	ISIC 2019	84.67%	84.3%	84.7%	0.95
Proposed Model	ISIC 2020	98.57%	98.78%	98.18%	1
Proposed Model	PAD-UFES-20	90.87%	90.52%	90.87%	0.95

et al. [20], who reported a slightly lower accuracy of 94% but a very high accuracy (99.24%), the proposed model maintains a better balance between precision and recall, which is crucial in multi-class classification tasks. Sarkar et al. [23], using Dempster–Shafer theory for fusion of multiple CNNs, achieved a low accuracy of 93.2%, while Vivek and Ramesh [37], despite the multimodal approach, achieved only an accuracy of 87.5%. These comparisons highlight that the proposed model maintains a more stable trade-off between sensitivity and specificity, strengthening its robustness for the HAM10000 dataset.

However, the accuracy of the proposed model on ISIC 2019 is lower at 84.67% with precision of 84.3% and recall of 84.7%. This result implies the difficulty posed by the class imbalance problem of this dataset, which is a common issue for ISIC 2019 as pointed out by Gessert et al. [18] who obtained 74.2% accuracy from a multimodal ensemble. Future work may include adjustable weightings, or better data augmentation methods to help with this problem.

Overall, the inclusion of tabular data, robust fusion using the Dempster–Shafer theory, and high AUC values emphasize the strength of the proposed multi-modal approach in effectively integrating visual and clinical data, setting a new benchmark in automated dermatological diagnosis.

Beyond performance metrics, the clinical value of the proposed multimodal fusion model lies in its potential to enhance diagnostic confidence and consistency in real-world dermatological practice. By integrating dermoscopic image data with structured clinical metadata, the model mimics the holistic reasoning process of experienced dermatologists, who routinely consider both visual patterns and contextual patient information. This multimodal approach is especially beneficial in ambiguous cases, where image-based predictions alone may be uncertain. Furthermore, the model's use of Dempster–Shafer fusion provides an interpretable framework for quantifying diagnostic uncertainty, which is essential in supporting clinical decision-making and risk stratification. Such features can aid less experienced clinicians in triaging complex skin lesions, reduce inter-observer variability, and potentially improve early detection rates of malignant cases, particularly in resource-limited settings where access to dermatology expertise is scarce. Ultimately, this work contributes not only to algorithmic performance, but also to the development of AI systems that are more aligned with real clinical workflows and decision support needs.

5.5. Clinical implications

Beyond its numerical performance, the proposed model offers tangible benefits for real-world dermatology practice. Its multimodal design, combining dermoscopic images with clinical data such as patient age, lesion location, and gender, aligns perfectly with the diagnostic process used by dermatologists [38,39]. By simulating human reasoning using visual and contextual cues, the model provides more generalized assessments, particularly useful for ambiguous or early-stage lesions. Moreover, its robustness on datasets with limited classes like PAD-UFES-20 and ISIC 2020 highlight its potential for deployment in primary care or teledermatology, where resources and labeled data are scarce. These clinical benefits underline not only the diagnostic accuracy of the model but also its usability for early detection and triage in real-world care workflows.

6. Limitations

The proposed model demonstrates strong performance in skin disease classification by leveraging multimodal evidence from dermoscopic images and clinical metadata. However, several limitations remain, offering directions for future work.

Firstly, classification performance on underrepresented classes, particularly in the ISIC 2019 dataset, was lower than dominant classes. This reflects the sensitivity of the model's belief mass generation and fusion process to data imbalance. Future work could incorporate class-aware loss functions (e.g., focal loss), advanced augmentation for minority classes, or reweighting strategies to improve robustness in imbalanced settings.

Secondly, although the discounted Dempster–Shafer fusion significantly improves decision reliability, it adds computational overhead from belief mass calculations and normalization. This may limit real-time inference or deployment on edge devices with constrained resources. Empirically, our fusion model requires approximately 11% more inference time compared to a baseline image-only EfficientNet, primarily due to additional attention layers, metadata processing, and the Dempster–Shafer combination logic. While this remains acceptable for most clinical use cases, lightweight alternatives, such as model pruning, quantization, or fusion approximations, could be explored to reduce this overhead.

Finally, interpretability remains critical in clinical contexts. While LIME and SHAP offer useful insights, future work could aim to make the fusion process itself more transparent, for example, by visualizing modality-specific trust factors α, or explaining how evidence was weighted across sources. This would enhance user trust and support clinical decision-making, improving adoption in practical settings.

7. Conclusion

This paper presents a multi-modal approach for skin disease classification, combining image features extracted by EfficientNet, tabular clinical data, and a discounted Dempster–Shafer fusion mechanism that incorporates trust-aware evidence weighting and uncertainty modeling. The results highlight the effectiveness of the proposed model, achieving a high accuracy of 98.57% on ISIC 2020, 90.87% on PAD-UFES-20, and 94.8% on HAM10000. Although the model showed relatively lower accuracy on ISIC 2019 due to class imbalance, it remained competitive across datasets and robust in multimodal integration.

Overall, this study demonstrates the potential of trust-weighted multi-modal learning to improve the accuracy, reliability, and interpretability of skin disease classification. It provides a foundation for future diagnostic systems that are both technically robust and clinically transparent.

CRediT authorship contribution statement

Amina Aboulmira: Writing – original draft, Visualization, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Hamid Hrimech:** Writing – review & editing, Validation, Project administration, Methodology, Investigation, Funding acquisition. **Mohamed Lachgar:** Writing – original draft, Validation, Supervision, Formal analysis, Conceptualization. **Hamza Malyana:** Conceptualization. **Khalid Lemouden:** Conceptualization.

Ethics approval

This research does not involve any studies with human participants, human tissue, or personal data. All datasets used are publicly available, anonymized, and do not require institutional ethical approval. Therefore, an ethics statement is not applicable to this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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